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## RESEARCH ARTICLE

# Improving Multi-Label Emotion Classification on Imbalanced Social Media Data With BERT and Clipped Asymmetric Loss

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**ABSTRACT** This research addresses the challenge of multi-label emotion classification on imbalanced datasets using a BERT-based model. Emotion classification, essential for applications like social media analysis and sentiment monitoring, often suffers from class imbalance, which hinders the detection of rare emotions. To address this, our model incorporates a clipped asymmetric loss function to prioritize minority classes while mitigating the dominance of frequent classes. We conducted extensive experimentation on GoEmotions and SemEval-2018 Task 1C datasets to demonstrate the model's effectiveness in achieving improved precision, recall, and F1-scores across various taxonomies, including GoEmotions, Ekman, and sentiment-grouped levels. Our approach achieved a notable improvement in macro-average F1-scores, increasing from 0.46 (baseline) to 0.54 on the GoEmotions dataset and 0.59 on the SemEval-2018 dataset. The results indicate significant advancements over standard BERT implementations and state-of-the-art models, particularly in recognizing rare emotions, making the model a robust solution for real-world, multi-label emotion classification tasks under imbalanced settings.

**INDEX TERMS** Asymmetric loss, BERT, emotion recognition, GoEmotions dataset, imbalanced data, multi-label classification, NLP.

## I. INTRODUCTION

Social media platforms have become a pivotal element of modern communication, allowing users to share their emotions through various media, including text, images, audio, and video. According to Kepios analysis, as of October 2024, social media users have reached approximately 5.22 billion globally, which represents about 63.8% of the global population [1]. Among the leading platforms, Facebook has more than 3 billion monthly active users who are actively engaged in sharing, commenting, and messaging [2]. YouTube has over 2.7 billion active users, with 1 billion watch hours daily and video uploads exceeding 500 hours per minute [3]. Instagram reports 500 million daily user stories and 95 million photos and videos uploaded [4]. Twitter (X) processes around 6,000 tweets every second,

resulting in approximately 200 billion tweets per year [5]. This widespread usage of social media emphasizes its dominance in communication, entertainment, and business operations [6]. Organizations and businesses are increasingly relying on the analysis of this data to understand public sentiment and make informed decisions [7], [8].

The proliferation of social media has fueled the emergence of emotion recognition as a significant area of research in Natural Language Processing (NLP) [9], [10], [11]. Emotion recognition goes beyond traditional sentiment analysis, where polarity is the only focus, to identify specific emotional states, such as love, joy, anger, sadness, and fear [12]. This field has diverse applications in multiple domains, such as healthcare [13], [14] mental health monitoring [15], education [16], customer service [17], [18], market research [19], and human-computer interaction [20]. Analyzing emotional trends on platforms like Facebook and Twitter provides a valuable tool for understanding public sentiment [21], aids in

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formulating effective strategies for crisis management [22], and facilitates the observation of how people react to events in real time [23], [24].

In this work, we focus on Textual Emotion Recognition (TER), a domain focused on identifying emotions conveyed in text through automatic methods. This domain has undergone significant progress, driven by the advancements in deep learning techniques. Researchers have developed sophisticated neural network architectures, like Semantic-Emotion Neural Network (SENN) [25] to improve emotion classification, particularly in text sourced from customer reviews and social media. The emergence of transformer-based models further revolutionized this domain by leveraging attention mechanisms to effectively capture contextual relationships in text, leading to an improved understanding of emotional expressions [26].

Due to the complex and varied nature of emotions and language, emotion classification still faces challenges. Multi-label classification, where text includes overlapping emotions, makes this task even more complex. To predict these emotions accurately, we require advanced models and methods [27]. The problem gets worse with imbalanced data, where emotions like joy and sadness are common, but others, like surprise or disgust, are rare. Due to this imbalance, models become biased and often fail to predict less frequent emotions correctly [28]. Common loss functions like cross-entropy do not handle this issue effectively. They focus too much on frequent emotions and ignore the rare ones. They often give misleading accuracy scores that hide poor performance on minority emotions [29].

Asymmetric loss function is an effective tool to address the problem of class imbalance. Unlike traditional loss functions, asymmetric loss adaptively adjusts the weights assigned to positive and negative samples during training. This prioritization ensures better handling of minority classes and reduces the model's bias towards overrepresented emotions [30], [31], [32]. This method improves the ability of the model to recognize rare emotions, achieving balanced performance on all classes. It ensures the model's reliability for real-world applications like social media monitoring and analyzing customers.

This paper presents a novel approach to multi-label emotion classification on imbalanced datasets using a BERT-based model with a custom Asymmetric Loss (ASL) function. The key contributions are:

We extend the ASL framework, originally designed for computer vision tasks [33], to the domain of Natural Language Processing (NLP), specifically for multi-label emotion classification. Our formulation retains the core principle of asymmetric focusing from the original loss function by Ridnik et al. while simplifying its implementation. The asymmetry is directly controlled by parameters, positive focusing ( $\gamma_{pos}$ ) and negative focusing ( $\gamma_{neg}$ ), instead of dynamically adjusting probability margins. Additionally, we incorporate explicit clipping of predicted probabilities to

ensure numerical stability and avoid the effects of extreme logarithmic values.

Our research directly addresses the significant issue of class imbalance prevalent in multi-label emotion datasets. The loss function prioritizes minority classes and ensures better performance for rare emotions without sacrificing accuracy for more common emotions.

By integrating the custom ASL with BERT, we leverage the contextual understanding and the representational power of transformer-based language models to enhance classification performance. This combination is shown to enhance the model's F1-scores significantly across multiple emotion taxonomies.

Extensive experiments demonstrate the robustness of our approach across multiple datasets (GoEmotions, SemEval-2018 Task 1C) and emotion taxonomies. These includes fine-grained (GoEmotions taxonomy), intermediate (Ekman taxonomy), and broad (sentiment grouping) levels of classification, showcasing its adaptability and effectiveness in diverse settings.

This document is further structured into Sections II to VII, covering related works, methodology, experiments, results, limitations, and conclusions, respectively.

## II. RELATED WORKS

Emotion classification has evolved significantly with the development of transformer-based models, especially BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. [34]. The bidirectional architecture of BERT processes context from both directions within a sentence, which enables better understanding of textual meaning, making it particularly effective for emotion classification tasks. However, multi-label emotion classification poses unique challenges. This section provides a review of recent research on BERT-based models and other innovative techniques aimed at addressing the challenges of multi-label emotion classification, as well as the use of asymmetric loss functions to manage imbalanced datasets in multi-label classification tasks.

Demszky et al. [35] proposed the GoEmotions dataset, the largest human-annotated multi-label emotion dataset to date. It consists of 58,000 Reddit English comments, labeled as either neutral or belonging to one or more of 27 different emotions. With its imbalanced nature and the inclusion of fine-grained emotions, the dataset serves as an excellent benchmark for evaluating multi-label emotion classification models. BERT models trained on this dataset have been shown to be effective at detecting frequent emotions but fail to identify rare emotions, which highlights the need for enhanced architectures and loss functions to better address class imbalance.

A sequence-to-sequence emotion modeling method, introduced by Huang et al. [36], leverages a bidirectional decoder to implicitly capture correlations among emotions. This approach outperformed the traditional binary relevance and classifier chain approaches by modeling the co-occurrence

of multiple emotions. However, it fails to address class imbalance, which leads to reduced recall for minority emotions.

Fei et al. [37] introduced a capsule network integrated with a variational autoencoder (VAE), enhanced with topic identification, for multi-label emotion detection. This model captures latent topic information from text to identify features associated with emotions. Although it performs well in detecting overlapping emotions due to the ability of the capsule network to capture relationships between features, this approach struggles with datasets that exhibit highly skewed distributions.

Chochlakis et al. [38] further improved multi-label emotion recognition by incorporating masked language modeling into BERT and leveraging label correlations. By explicitly modeling emotion interdependencies, this method enhances the prediction of co-occurring emotions and demonstrates robust performance across multiple languages but lacks mechanisms to mitigate class imbalance.

Class imbalance is a critical challenge in multi-label emotion classification, where some emotions occur less frequently than others. Conventional resampling techniques, such as undersampling and oversampling, partially mitigate this issue but introduce risks like overfitting or information loss [28], [39]. Techniques like Easy Data Augmentation (EDA) and ProtAugment were shown to be useful for improving the representation of minority classes where there is limited data [40], [41]. Tang et al. [42] showed that using data augmentation along with ensemble learning can improve BERT's performance on code-switched text, resulting in high F1-scores for minority emotions.

Wang et al. [43] analyzed how modern LLMs like GPT-4 and RoBERTa performed on the GoEmotions dataset. They used data augmentation and fine-tuning to improve classification results. However, GPT-4 performed worse than the fine-tuned BERT model in a multi-label scenario, with issues like hallucination, over-labeling, and over-interpretation. On the other hand, task-specific models like BERT, when combined with data augmentation techniques such as ProtAugment, showed significant improvements. This highlights the need for customized approaches to overcome the limitations of general-purpose LLMs for specialized tasks like fine-grained emotion detection.

Ameer et al. [44] introduced two graph attention-based models: UCCA-GAT, which captures semantic meaning, and Dep-GAT, which focuses on syntactic structures. Their study highlights the value of combining meaning and structure in graph-based models, especially for handling complex and imbalanced emotion datasets.

Custom loss functions serve as an important tool for handling class imbalance in multi-label classification. Focal loss, introduced by Lin et al. [45], modifies the conventional cross-entropy loss with a modulating term to focus more on misclassified examples while reducing the effect of easier ones. Zhu et al. [46] applied focal loss in a Bi-GRU-based speech emotion recognition model to improve recall for

rare emotions in imbalanced datasets. To address multi-label emotion detection, Deng and Ren [47] adapted focal loss by adding a feature extraction module and a correlation module. El Koshiry et al. [48] proposed a deep learning-based approach for cyberbullying detection, utilizing pre-trained GloVe embeddings and a focal loss technique. Ridnik et al. [33] introduced Asymmetric Loss for multi-label classification (ASL), which reduces the influence of easy negatives to prevent them from dominating the training process. Huang et al. [49] proposed Asymmetric Polynomial Loss (APL) to improve the performance of underrepresented classes. Cui et al. [50] introduced a loss function that uses class frequency to give more weight to underrepresented classes to improve their performance during training.

Models like Seq2Emo, topic-enhanced capsule networks, graph attention networks, and label correlation modeling have helped to manage overlapping emotions. Loss functions like focal loss and asymmetric loss have addressed imbalance issues. BERT-based models have shown significant progress in multi-label emotion detection. However, challenges like class imbalance and emotion overlap still exist. This study is built on these advancements by combining a BERT-based model with an asymmetric loss function to better handle multi-label classification and class imbalance.

### III. METHODOLOGY

In this section, the datasets are introduced, the methodology is discussed, and the experimental setup utilized in the research is detailed.

#### A. DATASETS

In our study, we carried out extensive experimentation on the GoEmotions dataset. We also tested our model using the SemEval-2018 Task 1C dataset to evaluate how well it performs on different datasets. Details about each dataset are given below.

##### 1) GoEmotions DATASET

GoEmotions is a large multi-label dataset with 58,000 Reddit comments that have been manually labeled with 27 emotions and a neutral category. It is ideal for fine-grained emotion classification but comes with challenges like class imbalance and overlapping emotions, making it a good benchmark for emotion classification models. As shown in Figure 1, the emotion label distribution within the dataset highlights the imbalance. Common emotions like admiration and gratitude appear most often, while less common emotions like grief and pride are underrepresented.

The neutral label, which appears 17,772 times, was excluded from the chart to better visualize the distribution of emotions. Since it is the most frequent category, its inclusion would make it difficult to compare the other emotions. However, neutral is still part of the dataset, as it helps the model distinguish between emotional and non-emotional content.

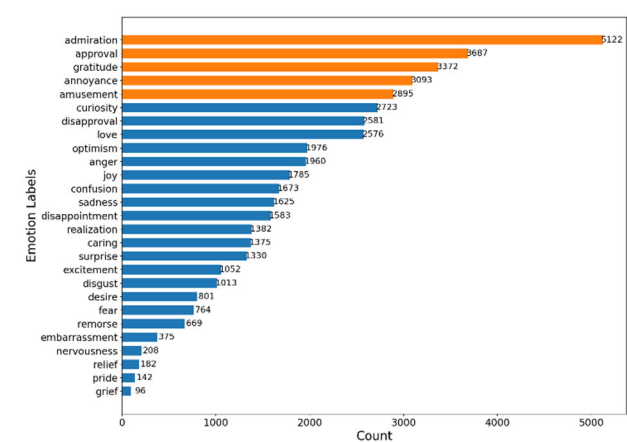


FIGURE 1. Emotion label distribution in the GoEmotions dataset (neutral label excluded).

We assessed the performance of our model using the three-level hierarchical taxonomy proposed by Demszyk et al., to categorize emotions at different levels of abstraction. At the sentiment level, the emotions are categorized into four broad categories as shown in Table 1. The Ekman level [51] further organizes the emotions into six groups plus neutral, as shown in Table 2. The GoEmotions taxonomy retains all 27 emotions plus a neutral label, offering the most fine-grained view of emotions. These hierarchical levels allowed us to evaluate the model’s performance at varying levels of abstraction, from broad sentiments to detailed emotion labels.

TABLE 1. Mapping of sentiment groups to GoEmotions labels.

| Sentiment Group | Mapped GoEmotions Labels                                                                                          |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Positive        | Admiration, Amusement, Approval, Caring, Desire, Excitement, Gratitude, Happiness, Love, Optimism, Pride, Relief  |
| Negative        | Anger, Annoyance, Disapproval, Disgust, Embarrassment, Fear, Grief, Nervousness, Remorse, Sadness, Disappointment |
| Ambiguous       | Confusion, Surprise                                                                                               |
| Neutral         | Neutral                                                                                                           |

TABLE 2. Mapping of Ekman groups to GoEmotions labels.

| Ekman Group | Mapped GoEmotions Labels                                                                                         |
|-------------|------------------------------------------------------------------------------------------------------------------|
| Anger       | Anger, Annoyance, Disapproval                                                                                    |
| Disgust     | Disgust                                                                                                          |
| Fear        | Fear, Nervousness                                                                                                |
| Joy         | Admiration, Amusement, Approval, Caring, Desire, Excitement, Gratitude, Happiness, Love, Optimism, Pride, Relief |
| Sadness     | Sadness, Disappointment, Embarrassment, Grief, Remorse                                                           |
| Surprise    | Surprise, Confusion                                                                                              |
| Neutral     | Neutral                                                                                                          |

2) SemEval-2018-TASK 1C DATASET

This is part of the Affect in Tweets dataset, which includes tweets gathered between 2016 and 2017. Designed for Task

E-c, it focuses on identifying multi-label emotions. While the dataset is available in Arabic, English, and Spanish, this study focuses only on the English subset. The dataset contains 10,983 tweets in the English subset, annotated with 11 distinct emotions: anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, and trust.

The label distribution of the English subset is shown in Figure 2, which highlights a significant imbalance in the representation of different emotions. Some emotions, such as joy, anger, and anticipation, are highly represented in the dataset, while others like trust and surprise appear far less frequently. This imbalance makes emotion detection models biased toward frequent emotions, often missing the rare ones. The multi-label nature of the tweets adds an additional layer of complexity compared to the traditional single-label classification tasks. This dataset serves as an important benchmark for developing and testing emotion recognition models, particularly in the context of multi-label emotion classification.

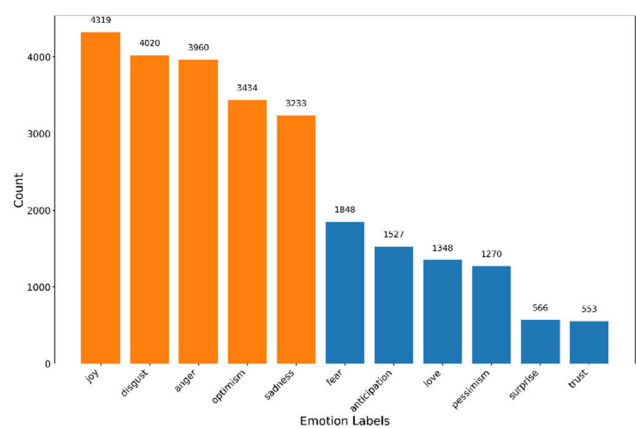


FIGURE 2. Emotion label distribution in the SemEval-2018 Task 1C dataset.

B. MODEL ARCHITECTURE

1) BERT MODEL

This approach is built on BERT (Bidirectional Encoder Representations from Transformers), which excels at capturing the contextual relationships between words and sentences. BERT’s bidirectional learning makes it an excellent choice for emotion detection, where context significantly influences the sentiment of a word. This work employs the BERT-base model, which includes an encoder consisting of 12 Transformer blocks. Each block includes a 12-head self-attention mechanism along with a hidden layer with 768 dimensions, resulting in a model comprising 110 million parameters. This architecture allows the model to process input sequences up to 512 tokens long and produce rich vector representations that encapsulate the meaning of the text. The BERT model has already been trained on large datasets like BookCorpus and Wikipedia. These pre-trained weights give the model a strong understanding of language, which we fine-tune further using the GoEmotions dataset to specialize it for emotion detection.



## 2) INPUT REPRESENTATION

To prepare the text data for BERT, we tokenize each comment using BERT's tokenizer, which divides the text into subwords and converts words into numeric token IDs that the model can process. Each text input is prefixed with a [CLS] token, which BERT uses to represent the entire sequence for classification purposes, and ends with a [SEP] token to signal the end of the text. To ensure consistent input dimensions, all text samples are either padded or truncated to a maximum sequence length of 128 tokens. Padding ensures all inputs are the same size, while truncation keeps things efficient without losing important information from most comments.

## 3) CLASSIFICATION LAYERS

After processing the input with BERT, the embedding derived from the [CLS] token, which represents the whole input text, is utilized. This embedding is then passed through a dense layer consisting of 512 units and activated by a ReLU function. This layer helps the model to recognize complex patterns and relationships within the features generated by BERT. Next, the output goes to the final layer, where a sigmoid activation function is applied, with each node representing a distinct emotion label. This setup allows the model to predict emotions independently, facilitating multi-label classification because it lets the model assign multiple emotions to the same input, which is important for overlapping emotions.

## 4) ASYMMETRIC LOSS FUNCTION

To effectively manage the class imbalance in our dataset, we integrated a custom-designed asymmetric loss function, adapted from the work of Ridnik et al. [33], into our model. This function is built on the concept of focal loss by emphasizing the learning of minority classes while controlling the influence of majority class samples.

This function addresses the challenge of class imbalance through adjustable parameters that modify how positive and negative samples contribute to the loss calculation. To prevent extreme values in predictions, which can lead to issues in logarithmic operations, we introduced clipping of predicted probabilities. The predicted probabilities ( $y_{pred}$ ) are clipped to a range defined by a small constant. This results in a clipped probability ( $\hat{y}$ ) as shown in (1).

$$\hat{y} = clip(y_{pred}, \varepsilon, 1 - \varepsilon) \quad (1)$$

where  $\varepsilon > 0$  is a small constant

This loss function defines two separate loss components: Positive Loss ( $L_{pos}$ ) and Negative Loss ( $L_{neg}$ ).

The positive loss focuses on samples where the true label  $y_{true} = 1$ . It emphasizes hard-to-classify positive examples while reducing the contribution of easy positive predictions. The formula for positive loss is shown in (2)

$$L_{pos} = -y_{true} \cdot \log(\hat{y}) \cdot (1 - \hat{y})^{\gamma_{pos}} \quad (2)$$

where

$y_{true}$  : True Label (1 for positive samples)

$\hat{y}$  : Predicted probability

$\gamma_{pos}$  : Positive focusing parameter that controls the degree of emphasis on positive samples.

The negative loss focuses on samples where the true label  $y_{true} = 0$ . It emphasizes hard-to-classify negative examples while decreasing the contribution of easy negative predictions. The formula for negative loss is given in (3)

$$L_{neg} = -(1 - y_{true}) \cdot \log(1 - \hat{y}) \cdot (\hat{y})^{\gamma_{neg}} \quad (3)$$

where

$1 - y_{true}$  : True Label (1 for negative samples)

$\hat{y}$  : Predicted probability

$\gamma_{neg}$  : Negative focusing parameter that controls the degree of emphasis on negative samples.

The total loss ( $L$ ) for each sample combines the positive loss and negative loss as shown in (4)

$$L = L_{pos} + L_{neg} \quad (4)$$

To compute the total loss ( $L_{total}$ ) for the dataset, the sample-wise loss is averaged, as shown in (5)

$$L_{total} = \frac{1}{N} \sum_{i=1}^N L_{pos}^{(i)} + L_{neg}^{(i)} \quad (5)$$

Figure 3 illustrates the architecture of the proposed BERT-based model which incorporates Clipped Asymmetric Loss (ASL) to handle class imbalance and improve performance. Our experimental findings validate the efficacy of the proposed asymmetric loss function in addressing class imbalance, demonstrating its ability to improve the recognition of minority classes while maintaining overall model performance.

## C. MULTI-LABEL EMOTION CLASSIFICATION PROCESS

The process for multi-label emotion classification is summarized in Algorithm 1. It describes the overall framework, from data preprocessing and model initialization to hyperparameter tuning and final evaluation. The training process integrates the custom-designed Clipped Asymmetric Loss function to address class imbalance. Algorithm 2 outlines the detailed steps for calculating the loss, emphasizing the separate handling of positive and negative samples.

## D. EXPERIMENTAL SETUP

### 1) DATA SPLITS AND STATISTICS

The GoEmotions and SemEval-2018 Task-1C datasets were split into training, validation, and test sets, according to the ratios suggested by their respective authors. For the GoEmotions dataset, approximately 75% of the data was allocated to the training set, with 9.4% assigned to both validation and test sets. This split ensures adequate training data for fine-grained emotion classification while providing balanced subsets for hyperparameter tuning and model evaluation [35]. The SemEval-2018 Task-1C dataset was

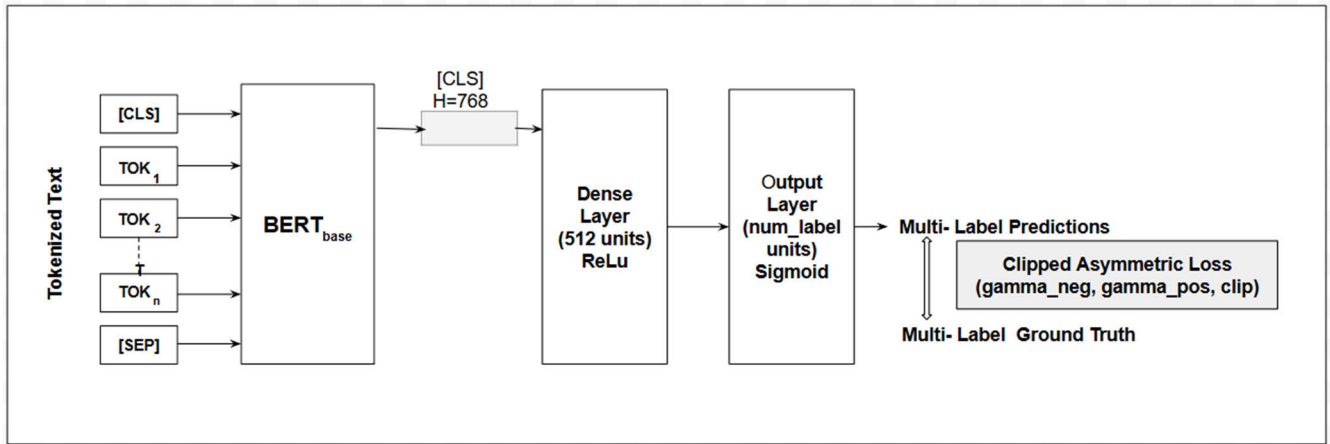


FIGURE 3. BERT with ASL: multi-label emotion classification architecture.

divided with approximately 62% allocated to the training, 8% to the validation, and 30% to the testing [52]. This ensures sufficient training data while retaining a large test set for evaluating the model on unseen data. Table 3 summarizes the dataset splits and the count of emotion labels available in each dataset.

TABLE 3. GoEmotions and SemEval-2018 Task-1C datasets summary.

| Dataset              | Train | Val  | Test | Total  | # of Emotion Labels |
|----------------------|-------|------|------|--------|---------------------|
| GoEmotions           | 43410 | 5426 | 5427 | 58000  | 27+ Neutral         |
| SemEval-2018 Task-1C | 6838  | 886  | 3259 | 10,983 | 11                  |

## 2) HYPERPARAMETER SETTINGS

The model was implemented in TensorFlow [53], using the Hugging Face Transformers library [54] to leverage BERT’s pre-trained architecture for tokenization, model loading, and optimization. Table 4 summarizes the hyperparameters used in this experiment. With a learning rate of  $2e-5$ , we used the Adam optimizer [55] in our experiments. Due to differences in dataset size and complexity, the GoEmotions dataset was trained with a batch size of 64, whereas a small batch size of 16 was chosen for the SemEval dataset. The positive focusing parameter ( $\gamma_{pos}$ ) and negative focusing parameter ( $\gamma_{neg}$ ) were set to 1 and 3, respectively, with a clipping constant ( $\epsilon$ ) of 0.04 for the GoEmotions dataset, and 1, 4, and 0.05 respectively, for the SemEval dataset. The model was trained on T4 GPU (Colab environment) [56] for 5 epochs on both datasets, with a training time of 180 seconds per step for the SemEval-2018 dataset and 417 seconds per step for the GoEmotions dataset. The best performance for sentiment-level and Ekman-level emotion classification was observed at epoch 2, after which the performance slightly declined.

Hyperparameters, including the batch size, focusing parameters ( $\gamma_{pos}$  and  $\gamma_{neg}$ ), clipping constant ( $\epsilon$ ), and learning

rates were optimized using a grid search strategy [57]. The grid search explored the following ranges:

- Batch Sizes: {8, 16, 32, 64}
- Positive Focusing Parameter ( $\gamma_{pos}$ ): {1, 2, 3, 4}
- Negative Focusing Parameter ( $\gamma_{neg}$ ): {1, 2, 3, 4}
- Clipping Constant ( $\epsilon$ ): {0.00, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07}
- Learning Rates: {1e-5, 2e-5, 5e-5}

These ranges were chosen based on prior studies and empirical tuning to balance computational cost and performance. The best configurations were determined based on the macro F1-score on the validation set.

TABLE 4. Hyperparameter settings.

| Parameter                   | Value (GoEmotions Dataset) | Value (SemEval-2018 Dataset) |
|-----------------------------|----------------------------|------------------------------|
| Learning Rate               | $2e-5$                     | $2e-5$                       |
| Batch Size                  | 64                         | 16                           |
| Positive Focusing Parameter | 1                          | 1                            |
| Negative Focusing Parameter | 3                          | 4                            |
| Clipping Constant           | 0.04                       | 0.05                         |
| Epochs                      | 5                          | 5                            |

## IV. RESULTS AND DISCUSSION

### A. MODEL PERFORMANCE ON GoEmotions TAXONOMY

The proposed model is tested across different taxonomies of the GoEmotions dataset and SemEval dataset to evaluate its robustness in emotion classification. To assess performance across emotion classes, standard evaluation metrics such as precision, recall, and F1-score are utilized. The results highlight its performance improvements in classifying emotions in highly imbalanced datasets.

**Algorithm 1** Multi-Label Emotion Classification

---

Input:  
 D\_train: Training dataset  
 D\_val: Validation dataset  
 D\_test: Test dataset  
 M: BERT Model  
 $\alpha$ : Learning rate

Output:  
 Trained Model M and evaluation metrics

Step 1: Initialize Model  
 $M \leftarrow$   
 LoadPretrainedBERTModelWithClassificationHead()

Step 2: Hyperparameter Tuning  
 For each combination of hyperparameters h in H do:  
   CompileModel(M, optimizer=h.optimizer,  
   learning\_rate=h.learning\_rate, loss=h.loss\_function)

Step 3: Training with Asymmetric Loss  
 Repeat for each epoch (1 to h.num\_epochs):  
   For each mini-batch in D\_train (batched and preprocessed) do:  
   predictions  $\leftarrow$  M.forward(mini\_batch\_input)  
   loss L  $\leftarrow$  ComputeAsymmetricLoss(predictions,  
   mini\_batch\_labels, h.loss\_params)  
   M.backward(L)  
   M.updateParameters(optimizer=h.optimizer,  
   learning\_rate=h.learning\_rate)  
   End For  
 End Repeat

Step 4: Validation  
 For each sample s in D\_val do:  
   probabilities  $\leftarrow$  M.predict(s)  
   predictions  $\leftarrow$   
 ConvertProbabilitiesToMultiLabel(probabilities,  
 threshold=h.prediction\_threshold)  
   precision\_val, recall\_val, f1\_score\_val  $\leftarrow$   
 CalculateMetrics(predictions, true\_labels(s))  
 End For

Step 5: Model Selection  
 If f1\_score\_val > best\_f1\_score Then  
   best\_f1\_score  $\leftarrow$  f1\_score\_val  
   best\_model  $\leftarrow$  Copy(M)  
 End If

End For

Step 6: Final Evaluation on Test Set  
 For each sample s in D\_test do:  
   probabilities  $\leftarrow$  best\_model.predict(s)  
   predictions  $\leftarrow$   
 ConvertProbabilitiesToMultiLabel(probabilities, threshold=0.5)  
   precision\_test, recall\_test, f1\_score\_test  $\leftarrow$  CalculateMet-  
 rics(predictions, true\_labels(s))  
 End For

Step 7: Return the Best Model and Test Evaluation Results  
 Return best\_model, {precision\_test, recall\_test, f1\_score\_test}

---

End Algorithm

Table 5 provides a detailed breakdown of the performance for each emotion class, including micro, macro, and weighted metrics. The proposed model demonstrates competitive performance across the 27 emotion classes in the GoEmotions taxonomy. It showed strong performance for frequent emotions such as gratitude (F1: 0.93) and

**Algorithm 2** Clipped Asymmetric Loss Calculation

---

Input:  
 y\_true: Ground truth labels (multi-label binary annotations)  
 y\_pred: Predicted probabilities for each label  
 $\gamma_{neg}$ : Focusing parameter for negative samples  
 $\gamma_{pos}$ : Focusing parameter for positive samples  
 clip: Threshold to clip low predicted values (to avoid instability)

Output: Loss value for the batch

Step 1: Numerical Stability Adjustment  
 $\varepsilon \leftarrow$  SmallPositiveConstant()  
 $y_{pred} \leftarrow$  ClipValues(y\_pred, min=  $\varepsilon$ , max=1 -  $\varepsilon$ )

Step 2: Apply Clipping (if clip is set)  
 If clip is not None Then  
   For each value p in y\_pred do:  
   If p < clip Then  
   p  $\leftarrow$  clip  
   End If  
   End For  
 End If

Step 3: Calculate Positive and Negative Losses  
 pos\_loss  $\leftarrow$  0  
 neg\_loss  $\leftarrow$  0  
 For each label l in y\_true do:  
   pos\_loss  $\leftarrow$  pos\_loss + (-y\_true[l] \* Log(y\_pred[l]) \*  
   (1 - y\_pred[l]) $^{\gamma_{pos}}$ )  
   neg\_loss  $\leftarrow$  neg\_loss + -(1 - y\_true[l]) \* Log(1 -  
   y\_pred[l]) \* (y\_pred[l]) $^{\gamma_{neg}}$ )  
 End For

Step 4: Combine and Average Losses  
 total\_loss  $\leftarrow$  Mean(pos\_loss + neg\_loss)

---

Return total\_loss

amusement (F1: 0.82), where sufficient training data is available. The model also shows improvements in recognizing less common emotions like love (F1: 0.79) and fear (F1: 0.67), indicating the effectiveness of the asymmetric loss function in addressing class imbalance. However, it achieves only moderate performance for minority emotions such as grief (F1: 0.44) and relief (F1: 0.35), revealing some limitations in handling extremely rare emotions. The overall performance on the GoEmotions dataset achieves a micro-average F1-score of 0.59 and a macro-average F1-score of 0.54, reflecting the model's balanced ability to recognize both frequent and infrequent emotions. Figure 4 shows a bubble chart showing the relationship between emotion frequency and F1-score. To further validate multi-label performance, we computed the Jaccard index, achieving a Jaccard accuracy of 0.5582 on the GoEmotions dataset with the proposed clipped asymmetric loss. This value, exceeding our macro F1 of 0.54, suggests that the model performs reasonably well in multi-label emotion classification.

TABLE 5. Performance metrics for GoEmotions taxonomy.

| Emotion          | Precision | Recall | F1-Score |
|------------------|-----------|--------|----------|
| Admiration       | 0.61      | 0.81   | 0.70     |
| Amusement        | 0.76      | 0.88   | 0.82     |
| Anger            | 0.46      | 0.56   | 0.50     |
| Annoyance        | 0.37      | 0.38   | 0.37     |
| Approval         | 0.36      | 0.42   | 0.39     |
| Caring           | 0.56      | 0.44   | 0.49     |
| Confusion        | 0.34      | 0.59   | 0.43     |
| Curiosity        | 0.48      | 0.71   | 0.57     |
| Desire           | 0.54      | 0.53   | 0.54     |
| Disappointment   | 0.31      | 0.35   | 0.33     |
| Disapproval      | 0.36      | 0.57   | 0.44     |
| Disgust          | 0.53      | 0.53   | 0.53     |
| Embarrassment    | 0.62      | 0.41   | 0.49     |
| Excitement       | 0.40      | 0.48   | 0.44     |
| Fear             | 0.62      | 0.73   | 0.67     |
| Gratitude        | 0.95      | 0.91   | 0.93     |
| Grief            | 0.67      | 0.33   | 0.44     |
| Joy              | 0.57      | 0.69   | 0.62     |
| Love             | 0.80      | 0.77   | 0.79     |
| Nervousness      | 0.43      | 0.43   | 0.43     |
| Neutral          | 0.70      | 0.60   | 0.64     |
| Optimism         | 0.54      | 0.59   | 0.56     |
| Pride            | 0.75      | 0.38   | 0.50     |
| Realization      | 0.29      | 0.26   | 0.28     |
| Relief           | 0.50      | 0.27   | 0.35     |
| Remorse          | 0.57      | 0.88   | 0.69     |
| Sadness          | 0.54      | 0.58   | 0.56     |
| Surprise         | 0.50      | 0.58   | 0.54     |
| Micro average    | 0.56      | 0.61   | 0.59     |
| Macro average    | 0.54      | 0.56   | 0.54     |
| Weighted average | 0.58      | 0.61   | 0.59     |

We analyzed the effectiveness of our proposed model using the GoEmotions dataset by comparing it with the following baseline and other advanced models.

BERT original [35]: This model utilizes the original BERT architecture without any specialized techniques. It uses the standard cross-entropy loss function and serves as the baseline model for comparison.

BERT DDA [43]: This model incorporates a Duplicate Data Augmentation (DDA), which applies random manipulations like synonym replacement, word swapping and deletion to improve diversity in training data.

BERT EMBED [43]: The model uses contextualized word embeddings from BERT for data augmentation by replacing or injecting words based on their contextual relevance.

PROT Augmented [43]: This model leverages BART (Bidirectional and Auto-Regressive Transformers) to paraphrase original text

CARER BERT [43]: BERT fine-tuned on the CARER dataset, which focuses on six primary emotions, before being fine-tuned on the GoEmotions dataset to transfer domain-specific knowledge. We also compared our result against CARER BERT combined with various augmentation techniques, including DDA, BERT augmented, and PROT augmented.

Table 6 presents a comprehensive comparison of the performance of the proposed model against the baseline

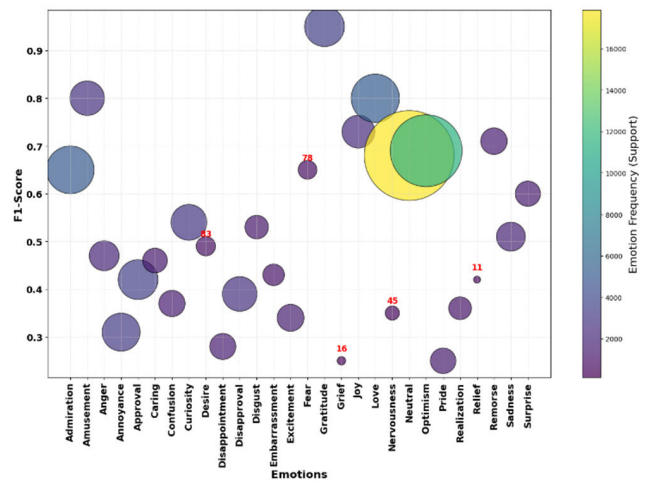


FIGURE 4. Emotion frequency vs F1-score – GoEmotions dataset.

model [35] and the other state-of-the-art models on the GoEmotions dataset. The proposed model outperforms the baseline and state-of-the-art models with a macro F1-score of 0.54, effectively balancing recall and precision. The results demonstrate the effectiveness of the asymmetric loss function incorporated in the proposed model in mitigating class imbalance and outperforming other models.

TABLE 6. Comparative results on GoEmotions dataset.

| Model                                        | Recall | Precision | F1 Macro    |
|----------------------------------------------|--------|-----------|-------------|
| Proposed Model                               | 0.56   | 0.54      | <b>0.54</b> |
| BERT original [35]                           | 0.40   | 0.63      | 0.46        |
| BERT DDA [43]                                | 0.50   | 0.58      | 0.51        |
| BERT EMBED [43]                              | 0.52   | 0.52      | 0.51        |
| BERT PROT Augmented [43]                     | 0.52   | 0.53      | 0.52        |
| CARER-BERT on original GoEmotions [43]       | 0.56   | 0.46      | 0.47        |
| CARER-BERT on DDA Augmented GoEmotions [43]  | 0.58   | 0.46      | 0.50        |
| CARER-BERT on BERT Augmented GoEmotions [43] | 0.58   | 0.47      | 0.51        |
| CARER-BERT on PROT Augmented GoEmotions [43] | 0.57   | 0.49      | 0.52        |

Figure 5 illustrates the F1-scores of the proposed model and the baseline model across 28 emotion categories from the GoEmotions dataset. The proposed model consistently outperforms the baseline, particularly for underrepresented emotions, where the asymmetric loss function helps address class imbalance. The neutral category is the only case where the proposed model slightly underperforms compared to the baseline, highlighting a potential area for further improvement.

The heatmap in Figure 6 visualizes the precision, recall, and F1-score for both the baseline and proposed models across four minority emotion classes: grief, relief, pride, and nervousness. It also highlights the improvements achieved by the proposed model. The proposed model achieves notable performance improvement in F1-score for all four categories, as evident from the heatmap, demonstrating its



effectiveness in addressing performance gaps, particularly for low-frequency emotions.

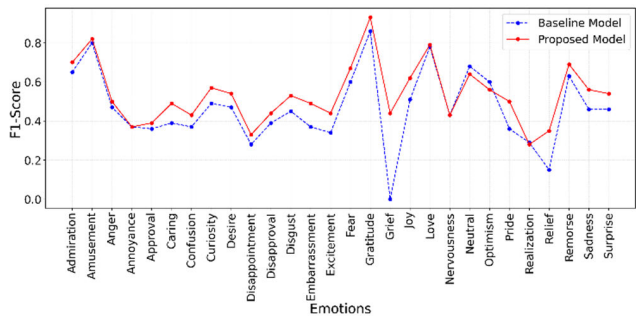


FIGURE 5. Comparison of F1-scores across emotions for the proposed and baseline models.

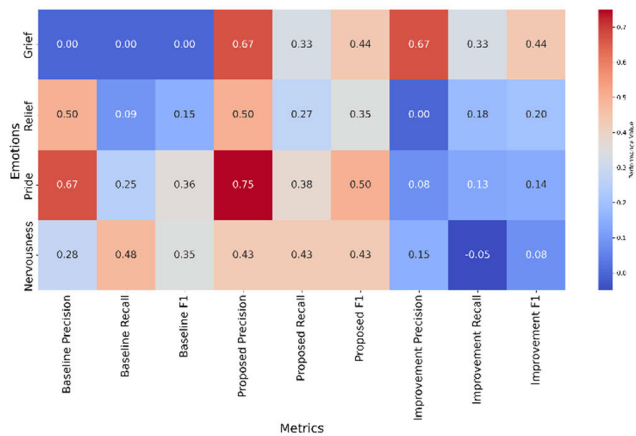


FIGURE 6. Heatmap of baseline vs proposed model performance across underrepresented emotions.

**B. MODEL PERFORMANCE ON EKMAN AND SENTIMENT LEVEL**

The proposed model demonstrates robust performance across both Ekman’s taxonomy of six primary emotions and sentiment grouping tasks. Table 7 provides a detailed summary of the results based on Ekman’s taxonomy. When mapped to Ekman’s taxonomy of six primary emotions plus neutral, the model achieved significant improvements. As shown in Table 7, the highest performance is observed for joy (F1: 0.82), highlighting the model’s strong ability to detect positive emotions. For negative emotions like fear (F1: 0.69) and anger (F1: 0.59), the model maintains reasonable recall while balancing sensitivity across classes. However, it struggles with rare emotions like disgust(F1:0.52), highlighting the challenge of class imbalance. The asymmetric function helps to mitigate the effects of class imbalance, which is evident in the higher recall for rare emotions like fear and disgust. Overall, the model achieves a macro-average F1-score of 0.65 and a micro-average F1-score of 0.71, demonstrating its ability to generalize on Ekman’s simpler taxonomy.

TABLE 7. Performance metrics on Ekman’s taxonomy.

| Emotion      | Precision | Recall | F1-Score |
|--------------|-----------|--------|----------|
| Anger        | 0.55      | 0.64   | 0.59     |
| Disgust      | 0.55      | 0.49   | 0.52     |
| Fear         | 0.65      | 0.73   | 0.69     |
| Joy          | 0.76      | 0.89   | 0.82     |
| Sadness      | 0.60      | 0.61   | 0.61     |
| Surprise     | 0.52      | 0.74   | 0.61     |
| Neutral      | 0.63      | 0.77   | 0.69     |
| micro avg    | 0.65      | 0.78   | 0.71     |
| macro avg    | 0.61      | 0.70   | 0.65     |
| weighted avg | 0.65      | 0.78   | 0.71     |

Table 8 provides a comparative analysis of the proposed model’s performance against the original BERT model and advanced graph-based approaches, Universal Conceptual Cognitive Annotation-Graph Attention Network (UCCA-GAT) and Dependency-Graph Attention Network (Dep-GAT), evaluated on the GoEmotions dataset using Ekman’s emotion taxonomy. The results illustrate that the proposed model achieved superior performance with the highest macro F1-score of 0.65 compared to other models.

TABLE 8. Comparative results on Ekman’s taxonomy.

| Model                    | Recall | Precision | F1 Macro    |
|--------------------------|--------|-----------|-------------|
| Proposed Model           | 0.61   | 0.70      | <b>0.65</b> |
| BERT original paper [35] | 0.59   | 0.69      | 0.64        |
| UCCA-GAT [44]            | 0.80   | 0.59      | 0.64        |
| Dep-GAT [44]             | 0.74   | 0.58      | 0.61        |

Figure 7 visualizes the comparative performance of the proposed model across key metrics using Ekman’s emotion taxonomy, showcasing how the proposed model balances performance across all metrics.

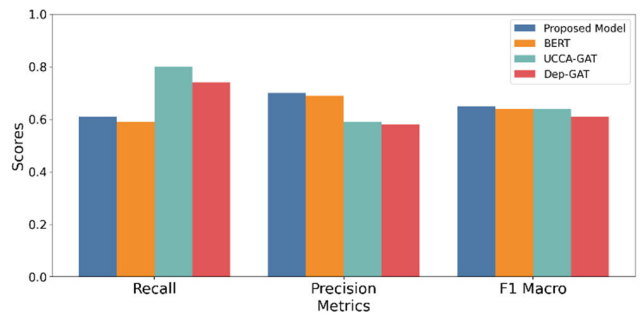


FIGURE 7. Model’s performance comparison on Ekman taxonomy.

Table 9 presents a detailed summary of the results based on sentiment grouping. The model achieves the highest performance for the positive sentiment group(F1:0.83). The model also achieves reasonable performance for the negative sentiment group(F1:0.69) and consistent performance across the neutral sentiment group(F1:0.68). Though the model poses a challenge in handling ambiguous class (F1:0.62),

the overall results reflect the robustness of the model. Attaining values of 0.72 and 0.70 for the micro-average and macro-average F1-scores, respectively, further highlights its balanced performance in these broader categories.

TABLE 9. Performance metrics on sentiment grouped data.

| Emotion          | Precision | Recall | F1-Score |
|------------------|-----------|--------|----------|
| Positive         | 0.79      | 0.87   | 0.83     |
| Negative         | 0.57      | 0.87   | 0.69     |
| Ambiguous        | 0.53      | 0.73   | 0.62     |
| Neutral          | 0.65      | 0.71   | 0.68     |
| Micro average    | 0.66      | 0.80   | 0.72     |
| Macro average    | 0.63      | 0.80   | 0.70     |
| Weighted average | 0.67      | 0.80   | 0.73     |

Table 10 compares the performance of the proposed model to original BERT implementation, while Figure 8 provides the comparative visualization.

TABLE 10. Comparative results on sentiment grouping.

| Model                    | Recall | Precision | F1 Macro |
|--------------------------|--------|-----------|----------|
| Proposed Model           | 0.63   | 0.80      | 0.70     |
| BERT original paper [35] | 0.65   | 0.74      | 0.69     |

Figure 9 illustrates how the proposed model performs compared to the baseline model across three emotion taxonomies. The results highlight that the proposed model adapts well to various taxonomies while ensuring balanced performance.

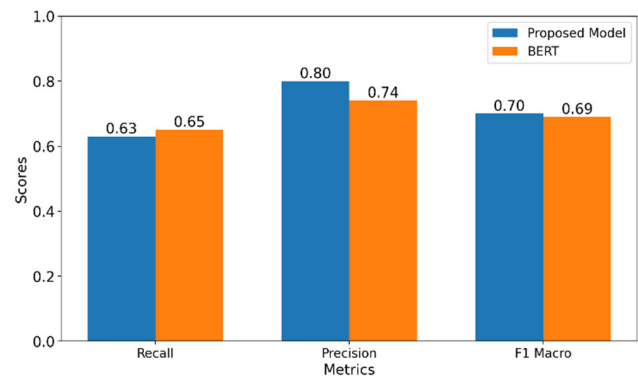


FIGURE 8. Model’s performance comparison on sentiment group.

C. MODEL PERFORMANCE ON SemEval-2018 DATASET

The model was further tested on the SemEval-2018 Task-1C dataset, which includes 11 distinct emotions in a multi-label setting. Table 11 shows the performance of each emotion, including, micro, macro, and weighted scores. On this dataset, the model achieved values of 0.70 for micro-average F1-score and 0.59 for macro-average F1-score, outperforming several advanced models. The model achieved

a Jaccard accuracy of 0.5822, indicating its reasonable performance on a different dataset. These results validate the model’s generalizability across datasets with varying complexities and class distributions.

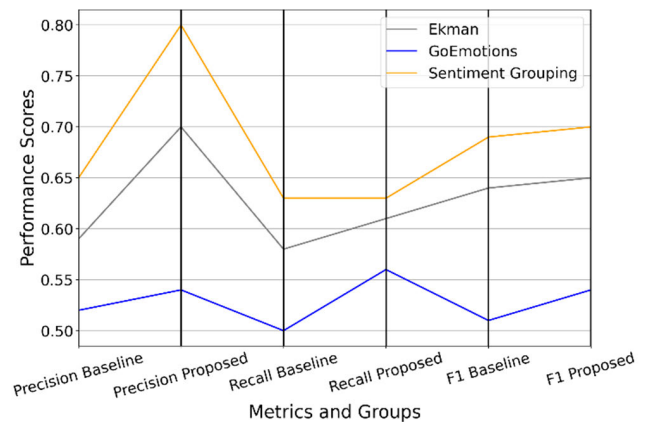


FIGURE 9. Model’s performance comparison across emotion taxonomies.

TABLE 11. Performance metrics for SemEval dataset.

| Emotion          | Precision | Recall | F1-Score |
|------------------|-----------|--------|----------|
| Anger            | 0.71      | 0.89   | 0.79     |
| Anticipation     | 0.38      | 0.39   | 0.38     |
| Disgust          | 0.67      | 0.87   | 0.76     |
| Fear             | 0.69      | 0.79   | 0.74     |
| Joy              | 0.73      | 0.94   | 0.82     |
| Love             | 0.47      | 0.83   | 0.60     |
| Optimism         | 0.63      | 0.91   | 0.75     |
| Pessimism        | 0.42      | 0.38   | 0.40     |
| Sadness          | 0.66      | 0.72   | 0.69     |
| Surprise         | 0.34      | 0.28   | 0.30     |
| Trust            | 0.19      | 0.33   | 0.24     |
| Micro average    | 0.62      | 0.79   | 0.70     |
| Macro average    | 0.54      | 0.67   | 0.59     |
| Weighted average | 0.62      | 0.79   | 0.70     |

The bubble chart in Figure 10 shows the relationship between emotion frequency and F1-score for the SemEval dataset.

Performance evaluation of the proposed model alongside other advanced emotion classification models using the SemEval dataset is shown in Table 12. Achieving a macro F1-score of 0.59, the proposed model outperforms benchmarks such as SpanEmo (F1: 0.578), BERTGCN (F1: 0.563), and BERT+DK (F1: 0.549). Compared to UCCA-GAT (F1: 0.60), the model shows comparable results. Figure 11 provides the visualization of the comparison.

We conducted a comparative assessment of the proposed model with existing methods across multiple datasets and taxonomies. With the macro F1-score on the GoEmotions taxonomy, the model surpasses existing approaches like DDA (F1: 0.51), ProtAug (F1: 0.51), and CARER-BERT on PROT-augmented data (F1: 0.52). Under Ekman’s Taxonomy, the proposed model achieves comparable results with prior models like UCCA-GAT (F1: 0.64) and Dep-GAT

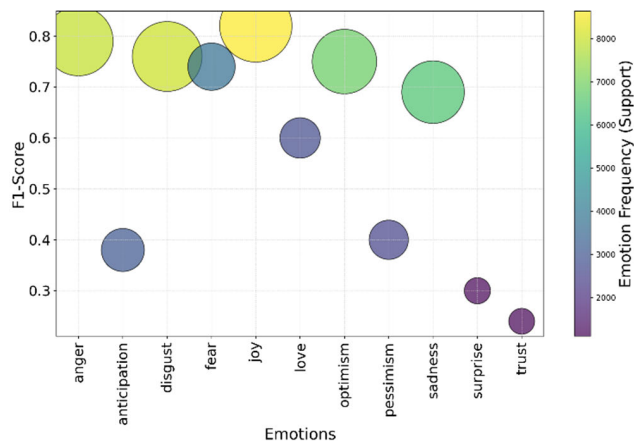


FIGURE 10. Emotion frequency vs F1-score - SemEval dataset.

TABLE 12. Comparative results on SemEval dataset.

| Model                       | Recall | Precision | F1 Macro    |
|-----------------------------|--------|-----------|-------------|
| Proposed Model              | 0.54   | 0.67      | <b>0.59</b> |
| UCCA-GAT [44]               | 0.659  | 0.62      | <b>0.60</b> |
| Dep-GAT [44]                | 0.683  | 0.524     | 0.518       |
| SpanEmo [58]                | -      | -         | 0.578       |
| BERT-GCN [59]               | -      | -         | 0.563       |
| BERT +DK [60]               | -      | -         | 0.549       |
| Transformer (Finetune) [61] | -      | -         | 0.561       |

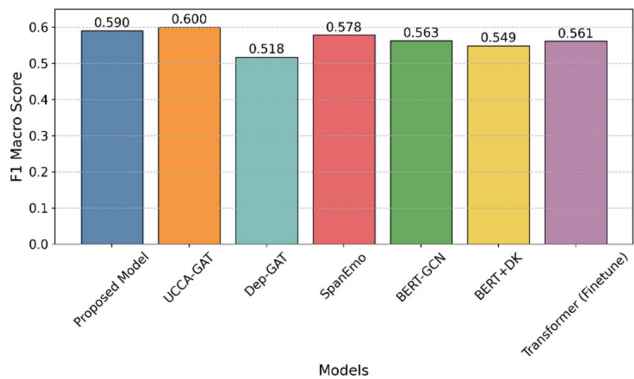


FIGURE 11. Model's performance comparison on SemEval dataset.

(F1: 0.61), emphasizing its ability to adapt to simpler taxonomies. The model achieves consistent improvements over baseline models, a macro F1-score of 0.70, and highlights its applicability for broader sentiment analysis tasks. When tested using the SemEval-2018 Task-1C dataset, the proposed model attained an impressive macro F1-score of 0.59, outperforming benchmarks such as SpanEmo (F1: 0.578), BERTGCN (F1: 0.563), and BERT+DK (F1: 0.549), and demonstrated comparable performance with UCCA-GAT (F1: 0.60), showcasing its efficacy in a competitive setting.

V. LIMITATIONS AND FUTURE DIRECTIONS

While our approach demonstrates significant advancements, challenges remain. First, the performance for extremely rare emotions like “relief” and “pride” is still limited by data sparsity. Further exploration of advanced data augmentation techniques, such as synthetic dataset generation, may be needed to address this limitation. Second, the current model does not consider multilingual data, restricting its applicability in global contexts. Future work could focus on adapting the framework to handle multi-language datasets. The high computational requirement of the model, particularly for large datasets, is a challenge for scalability. Employing optimization techniques like model pruning, quantization or knowledge distillation could be a choice to make the model more efficient.

VI. CONCLUSION

This study highlights the efficacy of a BERT-based model, augmented with a specialized asymmetric loss function, in addressing the complexities of imbalanced multi-label emotion classification. The asymmetric loss allows the model to focus on harder-to-learn minority classes, as evidenced by improved F1-scores. On the GoEmotions dataset, our approach elevates the macro F1-score from 0.46 (baseline BERT) to 0.54, a relative improvement of 17.4%, with notable gains in rare emotion recognition, such as a 50% increase in F1-score for fear (from 0.44 to 0.67). Consistent performance across GoEmotions (macro F1: 0.54), Ekman (macro F1: 0.65), and Sentiment Grouped Data (macro F1: 0.70) highlights the model’s adaptability to various emotion taxonomies. The comparative analysis further highlights the model’s competitiveness, achieving results on par with advanced methods. This demonstrates its robustness for practical applications such as monitoring social media and conducting sentiment analysis. Even though the model significantly improves the recognition of minority classes, challenges remain in detecting extremely rare emotions. Future work could include integrating the model with data augmentation techniques and refining the loss functions to further address the limitations. Additionally, extending the approach to handle multilingual datasets could enhance its applicability in global contexts. In conclusion, this research offers a promising solution for advancing emotion classification in natural language processing, especially in scenarios with inherent data imbalances, paving the way for more inclusive and accurate sentiment analysis systems.

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